

LESSON 5.8

DETERMINING DISTANCES



LESSON INTRODUCTION

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x-coordinate or the same y-coordinate, represented on the coordinate plane.

MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.

LEARNING TARGETS

- I can find distances between ordered pairs.
- I can solve a problem by plotting points on a coordinate plane and determining distances.

BENCHMARK COHERENCE

BUILDING ON

MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas

WORKING TOWARD

MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.

ESSENTIAL QUESTIONS

- How can you apply what you have learned about finding distance, area, and perimeter on the coordinate plane to solve real-world problems?

LESSON NARRATIVE

In this lesson, students apply their knowledge of finding distance, area, and perimeter on the coordinate plane to solve real-world problems. This lesson is a culmination of many topics in the entire unit and can be used to review and track student understanding of multiple topics involving the coordinate plane.

COMPONENT SUMMARY

5.8.1 WARM-UP (~5 MIN.)

Who is your math superhero?

5.8.3 EXPLORATION (10-15 MIN.)

Find the distance between two points and review methods to finding distance on the coordinate plane

5.8.5 YOUR TURN! (5 MIN.)

Find distance, perimeter, and area on a graph

5.8.2 GUIDED INSTRUCTION (10-15 MIN.)

Apply knowledge on distance to a race between two people

5.8.4 GUIDED PRACTICE (15 MIN.)

Find distance, perimeter, and area of a rectangle on the coordinate plane

5.8.6 WRAP-UP (~5 MIN.)

Find the missing vertex and solve for the area and perimeter on a graph

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Origin
- Absolute value
- Quadrant

MATERIALS

N/A

ADDITIONAL PREPARATION

- 5.8.4 Guided Practice contains an optional video to accompany the lesson.
- (Optional) Kinesthetic number line on the floor

TEACHER TIPS

COMMON MISCONCEPTIONS

- When solving problems with area, students may forget that they need to use square units.
- Students may confuse when to add the absolute values and when to subtract the absolute values when finding the distance between two points.
- Some students may count gridlines instead of spaces, which often leaves them with too many or too few.

THINGS TO CONSIDER

- Consider aligning on the definition of distance with students in this lesson, thinking of “distance” as “the number of units on the grid between two coordinates.”
- Consider providing sentence frames for students to explain their reasoning.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 5.8.2 Guided Instruction, students are prompted to look and think about what they have learned about distance and apply it to a real-world problem. While students are observing the comic, you can prompt them with a Notice and Wonder activity.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Throughout this lesson, students will be applying what they have learned about distance, area, and perimeter on the coordinate plane to real-world applications. Consider connecting 5.8.4 Guided Practice with language from MA.K12.MTR.7.1.

DIFFERENTIATE INSTRUCTION

This lesson incorporates a variety of activities to provide multiple entry points. 5.8.2 Guided Instruction may be accompanied by a Desmos Activity in which students can engage with the content through videos to increase student investment and engagement with the material. The videos provide additional visual entry points for students to consider the distance between points, while the digital coordinate plane allows students to practice graphing in an additional form.

5.8.1 WARM-UP: SUPERPOWERS

Component Overview: Students share their favorite superpower and superhero as a hook to make 5.8.2 Guided Instruction more engaging.

TEACHER MOVES

If students are unfamiliar with superheroes, prompt them to consider someone in their life who is a superhero to them or someone they know who is very strong in mathematics.

Example: Captain Marvel can fly faster than the speed of sound, which is approximately one mile in 4.7 seconds. Students can relate a variety of distances and identify how long it would take for Captain Marvel to fly that distance.



5.8.1 Warm-Up: Superpowers

1. If you could have a superpower, what would it be and why?

Answers will vary.

2. Think of a superhero whose powers relate to mathematics.

- a. Which superhero do you have in mind?

Answers will vary.

- b. Explain how his or her superpowers relate to mathematics.

Answers will vary.



5.8.2 Guided Instruction: The Flash vs. Superman

The Flash challenges Superman to a race. Who do you think will win?



Without watching the race, answer the following.

1. How can we find out the distance they covered?
Answers will vary. I can find the distance they covered without watching the race if I know where they started and where they ended the race.
2. How can we find out who won?
Answers will vary. Without watching the race, I could figure out who the winner is based on their speed.

Unfortunately, in the movie, *Justice League*, we were not able to see who wins. However, the Flash and Superman did race each other in the cartoon and comics version.

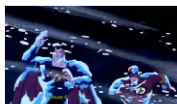
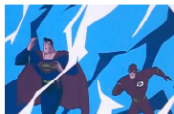
They raced through cities...



and on highways...



and through all kinds of conditions.



3. What do you see from the comic that is related to determining distance?
Answers will vary. I see red and blue lines. I see a starting point and an ending point.

5.8.2 GUIDED INSTRUCTION: THE FLASH VS. SUPERMAN

Component Overview: In this Guided Instruction, students consider distance in the context of a race and consider how they can determine the winner without watching the race. This activity can be accompanied by the Desmos Activity. If you decide not to have students complete the Desmos Activity, showing the video can provide a means of engagement for students.

ENRICHMENT

Encourage students to consider what information is needed to determine distance and who won. Ask scaffolding questions like:

- How can you determine distance on a graph? Can you use this to help determine distance in the real world?
- Without watching the race, what information do you need to determine the winner?

TEACHER MOVES

Notice and Wonder

While students are observing the comic, you can prompt them with a Notice and Wonder activity to help make connections between distance in the real world and distance on the coordinate plane, such as:

- What do you notice about the photos?
- How can that help you determine the distance?
- What do you see that could relate to distance?
- What do you still wonder?

5.8.3 EXPLORATION: LOOKING AT A TRAJECTORY

Component Overview: Students apply what they know about distance on the coordinate plane to find the distance between two points. Students review methods to finding distance when points are on the same side or different sides of the axis.

DIFFERENTIATE INSTRUCTION

The graphs provide visual entry points for students to engage with the material. Encourage students to understand how to use the graphs to calculate the distance and point out that the space between each gridline represents one mile.

TEACHER MOVES

Students may confuse when to add the absolute values and when to subtract the absolute values when finding the distance between two points. Encourage students to use the graphs to help make sense of when to add and when to subtract.

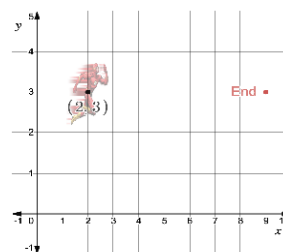
TEACHER MOVES

Students should know that if points are in the same quadrant, you can subtract the absolute value of the coordinates that the ordered pairs do not have in common. If points are in different quadrants, you can add the absolute values of the coordinates not in common.



5.8.3 Exploration: Looking at a Trajectory

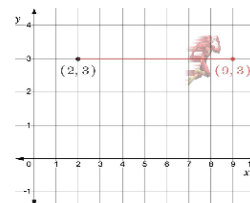
1. The following graph shows the Flash running from point $(2, 3)$.



What are the coordinates for where he ends?

The Flash ends at the point $(9, 3)$.

2. Consider the following graph.



- a. If each gridline on the x -axis represents 1 mile traveled, how many miles did the Flash travel?

$|9| - |2| = 7$ miles

- b. What is the distance between the starting and ending points for the Flash?

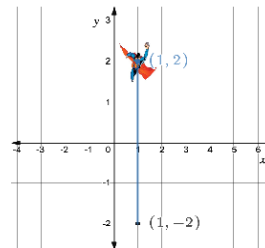
7 miles

- c. What do you notice between the coordinates of the starting point in relation to the coordinates at the end point?

Answers will vary. I noticed that the y -coordinates are the same, and the x -coordinate is 7 units larger at the endpoint.

3. When finding the horizontal distance between two points that lie on the same line, if the points are on the same side of the y -axis, find the ☐ sum ☒ difference of the absolute values between the x -values of the starting and ending points.

4. In the graph below, Superman starts flying at point $(1, -2)$. The y -axis represents the miles Superman travels.



- a. How many miles does he need to travel to get to point $(1, 2)$?

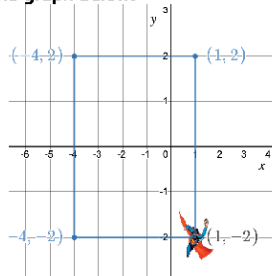
$|2| + |-2| = 4$ miles

- b. How do the coordinates of the starting point relate to the coordinates at the end point?

The starting and ending points have the same x -coordinate, but different y -coordinates.

5. When finding the vertical distance between two points that lie on the same line, if the points are on opposite sides of the x -axis, find the
☒ sum
☐ difference
 of the absolute values between the y -values of the starting and ending points.

6. Consider the graph below.



- a. The blue line shows the trajectory Superman followed starting from $(1, -2)$. Assuming the x - and y -axes represent miles traveled, explain how to determine the distance.
- Answers may vary. I notice that superman traveled in the shape of a rectangle. You could find the distance that Superman traveled by finding the perimeter of that rectangle.*
- b. What is the total distance traveled by Superman?
- Length of rectangle = $|2| + |-2| = 4$ miles
 Width of rectangle = $|-4| + |1| = 5$ miles
 Perimeter of rectangle = $2(4) + 2(5) = 18$ miles
 Superman traveled a total distance of 18 miles.*

5.8.3 EXPLORATION: LOOKING AT A TRAJECTORY (CONTINUED)

TEACHER MOVES

Think-Pair-Share

Leverage an instructional routine like Think-Pair-Share to give students an opportunity for sensemaking while refining their responses with a partner. Students are planning their approach for better fluency.

5.8.4 GUIDED PRACTICE: TOWN MAP ON A COORDINATE PLANE

Component Overview: Students name points on the coordinate plane with coordinates and then find the distance between those points. Students must add distances together to find the total distance traveled on the coordinate plane. Students then find the area and perimeter of a rectangle on the coordinate plane.

QUESTIONING

Consider using this activity as a model or exemplar for a project-based learning experience in which students demonstrate their mastery by mapping their neighborhood.

TEACHER MOVES

For question 1a, if students write:

- (2,-6), then review where the x- and y-coordinates go in an ordered pair (x,y).
- (6,2), then consider a review of positive and negative values on the coordinate plane.

ENRICHMENT

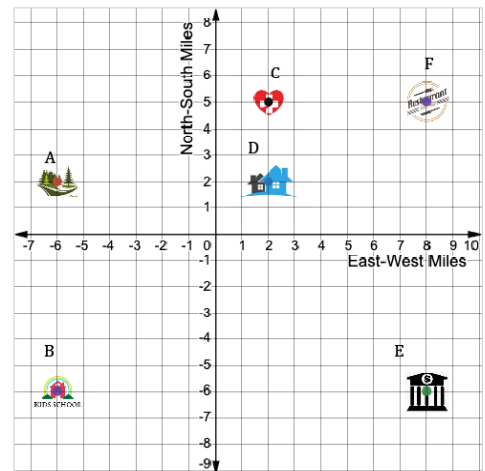
Notice that students need to find the total distance that Loki has traveled so far. Consider prompting students with questions like:

- How far was his most recent trip?
- How far had he already traveled beforehand?
- How far has he traveled altogether?



5.8.4 Guided Practice: Town Map on a Coordinate Plane

1. Consider the following graph.



a. What are the coordinates of the following places in town?

Park (A): (-6, 2)
 School (B): (-6, -6)
 Hospital (C): (2, 5)
 House (D): (2, 2)
 Bank (E): (8, -6)
 Restaurant (F): (8, 5)

b. Find the following distances.

The distance between the park at (-6, 2) and the school at (-6, -6) is 8 miles.

The distance between the hospital at (2, 5) and the house at (2, 2) is 3 miles.

The distance between the bank at (8, -6) and the school at (-6, -6) is 14 miles.

The distance between the restaurant at (8, 5) and the hospital at (2, 5) is 6 miles.

2. Loki leaves his house and heads towards the park. Then, he goes to school.

a. What distance has Loki travelled?
16 miles

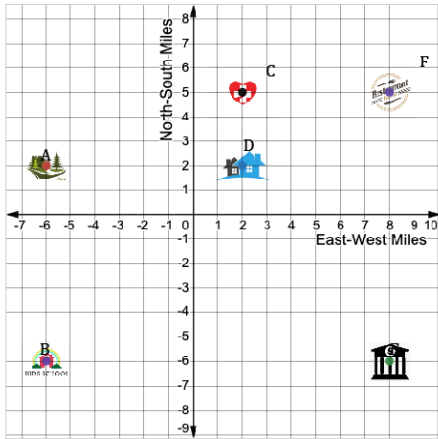
b. Loki gets out of school and goes to the bank to make a few transactions. Then, he heads toward the restaurant to have dinner. What distance has Loki traveled so far?

From bank to restaurant to dinner = 25 miles
Total distance = 25 + 16 = 41 miles

c. Before going home, Loki needs to stop at the hospital to visit a friend. After leaving the hospital, he finally goes home.

What distance did Loki travel altogether?
From restaurant to hospital to home = 9 miles
Total distance = 41 + 9 = 50 miles

3. The city commissioner will build a stadium so that it forms a square with the house, the park, and the school.



Explain where the stadium should be located.

Answers will vary. The stadium should be located at the point (2, -6). I know this because to make a square to make a square each side length has to be the same. So, the distance from the school and the house to the stadium are both 8 units.

4. Now that the stadium is built, what is the area and the perimeter formed by the stadium, the house, the park, and the school?
- Perimeter: 32 miles (mi.)
 - Area: 64 mi²

5.8.4 GUIDED PRACTICE: TOWN MAP ON A COORDINATE PLANE (CONTINUED)

TEACHER MOVES

Highlight student responses that specifically talk about the attributes of a square. Consider prompting students to think about what they know about squares to ensure they understand each side of a square has the same length, which will help students understand where to plot the stadium.

5.8.5 YOUR TURN!

Component Overview: Students find the distance, perimeter, and area of a soccer field using coordinates plotted on a graph.

QUESTIONING

Do you have access to your school's soccer field? Consider taking students outside to complete this activity.

ENRICHMENT

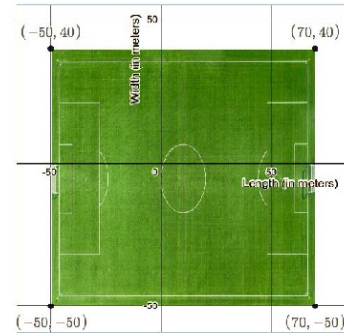
Consider providing students with scaffolding questions to help with finding the distance between two points.

- Are the coordinates on the same side or different sides of the axis?
- How do you find distance when on different sides of the axis?



5.8.5 Your Turn!

1. Consider the soccer field on the coordinate plane.



- a. Find the dimensions of the soccer field.

Length (horizontal): 120 meters (m)

Width (vertical): 90 m

- b. What is the perimeter, in meters, of the soccer field?

$$P = 2(120) + 2(90) = 240 + 180 = 420 \text{ m}$$

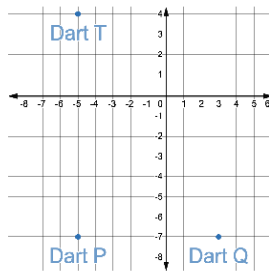
- c. What is the area, in square meters, of the soccer field?

$$A = 120 \times 90 = 10,800 \text{ m}^2$$



5.8.6 Wrap-Up: Ticket Out the Door

1. Zander is a competitive dart player. The graph shows a record of his last three darts.



- Plot the dart, Dart Z, where Zander should throw his next dart so that it forms a rectangle with the other three on the plane.
Dart Z would go at the point (3, 3).
- What is the perimeter of the rectangle formed by the darts?
 $P = 2(8) + 2(11) = 16 + 22 = 38 \text{ units}$
- What is the area of the rectangle formed by the darts?
 $A = 8 \times 11 = 88 \text{ units}^2$
- Zander will take back Darts P and Q. He wants to throw them so that they land on two points, forming a square with Darts T and Z. Where should the Darts P and Q land?
P (-5, -4) and Q (3, -4)

5.8.6 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Students apply what they've learned in this unit to find the last vertex of a rectangle and then determine its perimeter and area. Students then determine two more points to create squares.

DIFFERENTIATE INSTRUCTION

Notice the assessment provides summative data for plotting points, finding areas, and finding perimeter. Consider differentiating review for the end of the unit based on question 1.